Cell Factories and Mammalian Cell Culturing

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Cell Factories and Mammalian Cell Culturing (A13832)

Short Title: Cell Factories & Cell Culture
Faculty Science and Computing
Department Science
Cluster Biology
Author EMCNEELA
Credits 10 **Level:** Advanced

Description of Module / Aims

This module will investigate and compare the use of a variety of host cell systems, including microbial, insect, plant and mammalian, for the production of biopharmaceuticals. In particular, emphasis will be given to recent developments in the optimisation of biopharmaceutical production by host cells. The module will also provide a theoretical introduction to and direct practical training in the maintenance and culture of mammalian cells. Students will be directly trained in basic animal cell culturing methods and bioanalytical techniques particularly relevant to the biopharmaceutical industry.

Programmes

`MOLB-0003`	BSc (Hons) in Applied Biology with Quality Management (`WD_SQMBI_B`)	4 / 1 / M
`BIOL-0028`	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`)	4 / 7 / M

Indicative Content

- Bacterial expression systems for recombinant protein production; vectors and promoters; advantages and disadvantages of and new developments in bacterial expression systems
- Expression of recombinant proteins in yeast and filamentous fungi; expression vectors and promoters; post-translational modifications
- Insect expression systems; baculovirus expression vectors
- Expression of recombinant proteins in plants and plant cells; plant-expressed vaccines; new technologies
- Overview and history of mammalian cell culture; primary cells and continuous cell lines; mammalian cell growth and death; transformation and immortalization
- Expression of recombinant proteins for biopharmaceutical production in mammalian cells; mammalian cell transfection and expression vectors; cell line engineering; selection and screening of cell lines for biopharmaceutical production
- Principles of mammalian cell culture; culture conditions and cell culture media; generation and maintenance of cell banks; cell line characterisation and monitoring and contamination of cell cultures
- Scale-up of mammalian cell culture for biopharmaceutical production; bioreactors and scale-up using suspension cells and adherent cells
- Cell-based therapies; allogeneic and autologous cell-based therapies; stem cell research and potential applications

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Compare and contrast different eukaryotic and prokaryotic expression systems for recombinant proteins production.
2. Appraise recent developments in the use of insect, plant and animal cells for biopharmaceutical production and therapeutic applications.
3. Distinguish between the different bioreactor configurations used in the biopharmaceutical industry in relation to their modes of operation and their spheres of application.
4. Interpret the molecular mechanisms involved in mammalian cell division, cell growth and cell death.
5. Evaluate recent developments in the genetic engineering of cell lines and in process optimisation to enhance recombinant protein production.
6. Appraise the principles and methods of mammalian cell line characterisation and biosafety analysis from a quality control perspective.
7. Manage and maintain mammalian cell cultures, demonstrating skills in media preparation, sub-culturing and biocontamination control.
8. Prepare and complete tests using simple cell-based bioassays and bioanalytical techniques.

Learning and Teaching Methods

- This module will use a blended learning approach combining face-to-face lectures with e-learning available through the platform Moodle.
- Student engagement will be enhanced through use of the interactive e-learning platform, in-class discussions through workshops, webinars, research of current relevant literature and informative presentations.
- A learning and teaching strategy appropriate to level 8 will be adopted throughout with opportunities for formative assessment to encourage reflective, critical thinking.
- Practicals will be delivered to underpin and supplement theoretical material.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	60%		1,3,4,5,6
Continuous Assessment	40%		
Practical		30%	7,8
Presentation		10%	2

Assessment Criteria

<40%: No understanding of the basics of cell factories and mammalian cell culture. Unable to carry out critical thinking and analysis. Lack of competence in associated laboratory skills.

40%-49%: Able to understand basic concepts of cell factories and mammalian cell culture. Displays some ability in associated practical skills. Has difficulty with critical thinking and problem solving.

50%-59%: A good understanding of cell factories and mammalian cell culture with demonstration of some critical thinking and problem solving skills. Displays reasonable competence in associated laboratory skills.

60%-69%: A good understanding of cell factories and mammalian cell culture and good critical thinking and problem solving skills. Displays very good competence in associated laboratory skills.

70%-100%: Excellent in all areas. Demonstrates a highly developed and extensive comprehension of the subject manner. Shows ability to evaluate, debate and relate topics and shows evidence of extensive independent learning and knowledge of current literature and recent developments in the area. Displays excellent competence in associated laboratory skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	12	
Practical	36	
Independent Learning	186	

Essential Material

- Davis, J.M. _Animal Cell Culture: Essential Methods_. Hoboken, N.J.: John Wiley & Sons Inc., 2011.

Supplementary Material

- "American Association for the Advancement of Science." www.aaas.org
- "American Type Culture Collection." www.atcc.org
- "BioProcess International." http://www.bioprocessintl.com/
- "Gibco Cell Culture Basics." https://www.thermofisher.com/ie/en/home/references/gibco-cell-culture-basics.html
- Glick, B.R., J.J. Pasternak and C.L. Patten. _Molecular Biotechnology: Principles and Applications of Recombinant DNA_. 4th Ed. Washington: ASM Press, 2009.
- Smith, J.E. _Biotechnology_. Fifth Edition. UK: Cambridge University Press, 2009.

Requested Resources

- Room Type: Biology Lab
- Lecture Room: Loose Seated